



SCHOOL OF COMPUTER SCIENCE AND INFORMATION SYSTEMS
WINTER SEMESTER 2025-26
FINAL ASSESSMENT

April 15, 2026

Programme name: M. Tech Software Engineering (Integrated)
Course mode: Lab Based Component
Class number: VL2025260503869
Course: ISWE406P - AGILE Development process and Devops lab
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Registration number : 23MIS0296

Total Marks:50

SET – 5

ANSWER ALL THE QUESTION

Online Donation System

Problem Statement:

Ensure donations are processed and recorded accurately

Task:

1. develop donation processing logic
2. write junit test for transaction validation
3. implement ci/cd pipeline with automated testing
 - setup version control using github
 - configure ci pipeline (eg.jenkins/github action)to trigger on code commits
 - automate build process using maven
 - integrate automated unit test in pipeline
 - configure cd pipeline for automatic deployment to staging / production
4. deploy services on docker and kubernetes
 - containerize application using docker (write docker file)
 - build and push docker images to container registry (docker hub)
 - create kubernetes deployment and service yaml files

Answer:

1. Java program

Code:

Donation.java:

```
package com.donation;

public class Donation {
    private String donationId;
    private double amount;
    private String donorName;
    private boolean processed;

    public Donation(String donationId, double amount, String donorName) {
        this.donationId = donationId;
        this.amount = amount;
        this.donorName = donorName;
        this.processed = false;
    }

    public String getDonationId() {
        return donationId;
    }

    public double getAmount() {
        return amount;
    }

    public String getDonorName() {
        return donorName;
    }

    public boolean isProcessed() {
        return processed;
    }

    public void setProcessed(boolean processed) {
        this.processed = processed;
    }
}
```

DonationProcessor.java:

```
package com.donation;

public class DonationProcessor {

    public boolean processDonation(Donation donation) {
        if (donation == null) {
            return false;
        }

        if (donation.getAmount() <= 0) {
            return false;
        }

        if (donation.getDonorName() == null || donation.getDonorName().trim().isEmpty()) {
            return false;
        }
    }
}
```

```

    }

    donation.setProcessed(true);
    System.out.println("Donation Processed Successfully: ID=" + donation.getDonationId() +
        ", Amount=$" + donation.getAmount() +
        ", Donor=" + donation.getDonorName());

    return true;
}

public static void main(String[] args) {
    DonationProcessor processor = new DonationProcessor();
    Donation myDonation = new Donation("TXN-101", 50.0, "John Doe");

    if (processor.processDonation(myDonation)) {
        System.out.println("System initialized and handled first donation.");
    } else {
        System.out.println("Failed to process transaction.");
    }
}
}

```

2. Junit Testing

Code:

Output:

```

C:\Windows\System32\cmd.exe X + v
C:\Users\Sooriya\Desktop\vit\6_WINTER semester 2025-26\c_Aglie\FAT LAB>cd "c:\Users\Sooriya\Desktop\vit\6_WINTER semester 2025-26\c_Aglie\FAT LAB\donation-system"
c:\Users\Sooriya\Desktop\vit\6_WINTER semester 2025-26\c_Aglie\FAT LAB\donation-system>mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.donation:donation-system >-----
[INFO] Building donation-system 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO]
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ donation-system ---
[INFO] Deleting c:\Users\Sooriya\Desktop\vit\6_WINTER semester 2025-26\c_Aglie\FAT LAB\donation-system\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ donation-system ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory c:\Users\Sooriya\Desktop\vit\6_WINTER semester 2025-26\c_Aglie\FAT LAB\donation-system\src\main\resources
[INFO]
[INFO] --- compiler:3.10.0:compile (default-compile) @ donation-system ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 2 source files with javac [debug target 25] to target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ donation-system ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory c:\Users\Sooriya\Desktop\vit\6_WINTER semester 2025-26\c_Aglie\FAT LAB\donation-system\src\test\resources
[INFO]
[INFO] --- compiler:3.10.0:testCompile (default-testCompile) @ donation-system ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 25] to target\test-classes
[INFO]
[INFO] --- surefire:3.0.0-M5:test (default-test) @ donation-system ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.donation.DonationProcessorTest
Donation Processed Successfully: ID=TXN-001, Amount=$100.5, Donor=Alice Smith
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.166 s - in com.donation.DonationProcessorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ donation-system ---
[INFO] Building jar: c:\Users\Sooriya\Desktop\vit\6_WINTER semester 2025-26\c_Aglie\FAT LAB\donation-system\target\donation-system-1.0-SNAPSHOT.jar
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 8.892 s
[INFO] Finished at: 2026-04-15T18:10:44+05:30
[INFO]
[INFO] -----
c:\Users\Sooriya\Desktop\vit\6_WINTER semester 2025-26\c_Aglie\FAT LAB\donation-system>

```

3.CI/CD pipeline Creation and execution

Code:

```
pipeline {
    agent any

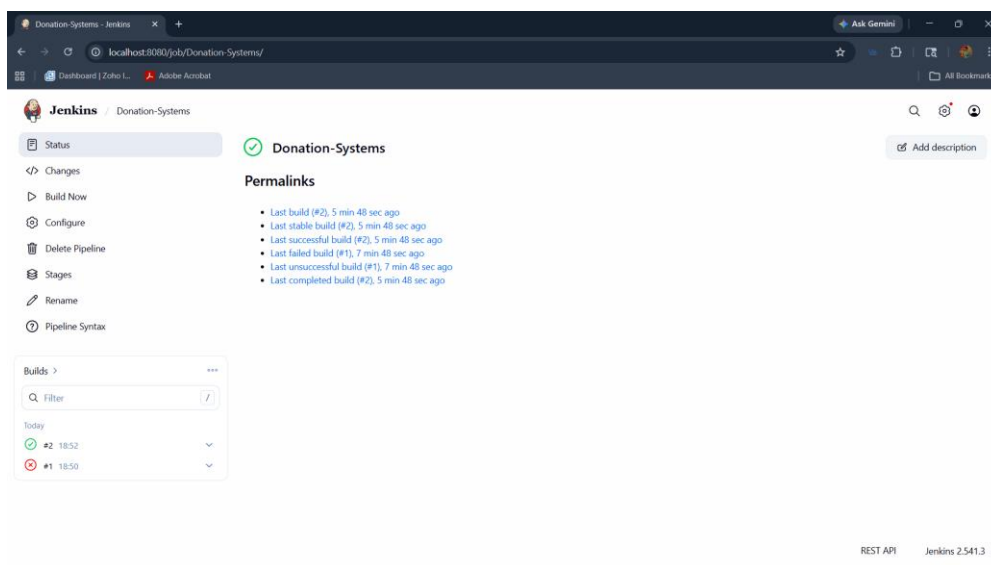
    stages {
        stage('Checkout Code') {
            steps {
                git branch: 'main', url: 'https://github.com/sooriya-moorthy-0107/AGILE-LABFAT-Online-Donation-System.git'
            }
        }

        stage('Build with Maven') {
            steps {
                bat 'C:\\apache-maven-3.9.14\\bin\\mvn.cmd clean package'
            }
        }

        stage('Docker Containerization') {
            steps {
                bat 'docker build -t sooriya/donation-system:latest .'
            }
        }

        stage('Deploy to Kubernetes') {
            steps {
                // Deploy using your specific user kubeconfig!
                bat 'kubectl --kubeconfig=C:\\Users\\Sooriya\\.kube\\config apply -f k8s/deployment.yaml'
                bat 'kubectl --kubeconfig=C:\\Users\\Sooriya\\.kube\\config apply -f k8s/service.yaml'
            }
        }
    }
}
```

Output:



The screenshot displays the Jenkins web interface for a pipeline named 'Donation-Systems'. The interface is viewed through a browser at the URL 'localhost:8080/job/Donation-Systems/'. On the left sidebar, the 'Status' tab is selected, showing options like 'Changes', 'Build Now', 'Configure', 'Delete Pipeline', 'Stages', 'Rename', and 'Pipeline Syntax'. The main content area shows the pipeline's status as 'Success' with a green checkmark. Below this, a 'Permalinks' section lists recent builds: 'Last build (#2), 5 min 48 sec ago', 'Last stable build (#2), 5 min 48 sec ago', 'Last successful build (#2), 5 min 48 sec ago', 'Last failed build (#1), 7 min 48 sec ago', 'Last unsuccessful build (#1), 7 min 48 sec ago', and 'Last completed build (#2), 5 min 48 sec ago'. A 'Builds' section at the bottom shows a list of builds with a filter input. The first build is marked as successful (green circle) and the second as failed (red circle). The footer of the page indicates 'REST API' and 'Jenkins 2.541.3'.

4. Deploy services on Docker and Kubernetes

Code:

Output:

```
c:\Users\Sooriya\Desktop\vit\6_WINTER semester 2025-26\c_Aglie\FAT LAB\donation-system>docker build -t sooriya/donation-system:latest .
[*] Building 3.6s (16/16) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 404B
=> [internal] load metadata for docker.io/library/eclipse-temurin:25-jre
=> [internal] load metadata for docker.io/library/maven:3.9-eclipse-temurin-25
=> [auth] library/eclipse-temurin:pull token for registry-1.docker.io
=> [auth] library/maven:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [stage-1 1/5] FROM docker.io/library/eclipse-temurin:25-jre@sha256:5da8c1d73846ea62dd635a7ee86aa3fd3c0bebed1aeca61fd57a5617fef4eb
=> resolve docker.io/library/eclipse-temurin:25-jre@sha256:5da8c1d73846ea62dd635a7ee86aa3fd3c0bebed1aeca61fd57a5617fef4eb
=> [build 1/5] FROM docker.io/library/maven:3.9-eclipse-temurin-25@sha256:c82963d26ac21d69ba333e28bd22c179cbef22b3561d67744a16342c8c33dc94
=> resolve docker.io/library/maven:3.9-eclipse-temurin-25@sha256:c82963d26ac21d69ba333e28bd22c179cbef22b3561d67744a16342c8c33dc94
=> [internal] load build context
=> => transferring context: 502B
=> CACHED [stage-1 2/5] WORKDIR /app
=> CACHED [build 2/5] WORKDIR /app
=> CACHED [build 3/5] COPY pom.xml .
=> CACHED [build 4/5] COPY src ./src
=> CACHED [build 5/5] RUN mvn clean package -DskipTests
=> CACHED [stage-1 3/3] COPY --from=build /app/target/donation-system-1.0-SNAPSHOT.jar /app/donation.jar
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:2a3aa41bf8fe0b3479344ced0384ddde13a8a48f6875f72fa98b1c3688707a46
=> => exporting config sha256:c952f9e9678d17250b14e3cb09e5d7ed5e6912b985b155ff62a7f013e53fbf64
=> => exporting attestation manifest sha256:c013dca5207b7404b092dc481bccdc7b481d83215524742927b1828493b4861d
=> => exporting manifest list sha256:486d055ba3947de0d4787e9d972f427680132d68e946827ab204c654cf43d2fb2
=> => naming to docker.io/sooriya/donation-system:latest
=> => unpacking to docker.io/sooriya/donation-system:latest
0.0s
```

```
c:\Users\Sooriya\Desktop\vit\6_WINTER semester 2025-26\c_Aglie\FAT LAB\donation-system>
```

```
c:\Users\Sooriya\Desktop\vit\6_WINTER semester 2025-26\c_Aglie\FAT LAB\donation-system>kubectl apply -f k8s/deployment.yaml
deployment.apps/donation-system-deployment configured
```

```
c:\Users\Sooriya\Desktop\vit\6_WINTER semester 2025-26\c_Aglie\FAT LAB\donation-system>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
donation-system-deployment-5ffcfffb74c-htksf   1/1     Running   0           4s
donation-system-deployment-5ffcfffb74c-x29wz    1/1     Running   0           5s
feedback-app-5c5f7859c7-d2tsr                 1/1     Running   3 (31m ago) 6d23h
feedback-app-5c5f7859c7-dv7s9                 1/1     Running   3 (31m ago) 6d23h
feedback-app-5c5f7859c7-wdiz6n                 1/1     Running   3 (31m ago) 6d23h
online-quizz-system-7bc656978d-8xtkh            1/1     Running   2 (31m ago) 6d19h
online-quizz-system-7bc656978d-ljtx8            1/1     Running   2 (31m ago) 6d19h
```

```
c:\Users\Sooriya\Desktop\vit\6_WINTER semester 2025-26\c_Aglie\FAT LAB\donation-system>
```